S5.6 Mathematics Mock Exam - Semester 2 2026

Solutions & Marking Scheme (Pedagogical Guide)

Topic 1: Quadratics and the Parabola

Question 1

Consider the quadratic function $y = 2x^2 - 12x + 10$.

Part a) Write in vertex form $y = a(x-p)^2 + q$.

Step-by-step Solution: 1. **Factor the leading coefficient ($a = 2$)** out of the terms containing x:

$$y = 2(x^2 - 6x) + 10$$

2. **Complete the square** inside the parentheses. To do this, take the coefficient of x (which is -6), divide it by 2 to get -3, and square it to get 9. We add and subtract 9 inside the parentheses to maintain equality:

$$y = 2(x^2 - 6x + 9 - 9) + 10$$

3. **Rewrite the perfect square trinomial** $(x^2 - 6x + 9)$ as a squared binomial $(x-3)^2$:

$$y = 2\left[(x-3)^2 - 9\right] + 10$$

4. **Distribute the 2** back and simplify the constant terms:

$$y = 2(x-3)^2 - 18 + 10$$
$$y = 2(x-3)^2 - 8$$

- **Marking Rubric (3 Marks total):**
  - **1 Mark**: Correctly factoring out $a = 2$.
  - **1 Mark**: Correctly completing the square inside the brackets.
  - **1 Mark**: Correct final vertex form $2(x-3)^2 - 8$.

Part b) State the vertex coordinates and the line of symmetry equation.

Step-by-step Solution: 1. The vertex form is $y = a(x-p)^2 + q$, where (p, q) is the vertex. * Here, $p = 3$ and $q = -8$. So, the **vertex coordinates** are $(3, -8)$. 2. The **line of symmetry** is the vertical line passing through the vertex $x = p$. * Therefore, the equation of the line of symmetry is $x = 3$.

- **Marking Rubric (2 Marks total):**
  - **1 Mark**: Vertex coordinates $(3, -8)$ written correctly.
  - **1 Mark**: Equation of line of symmetry written as $x = 3$ (must be an equation, not just the number 3).

Part c) Find the roots (zeros) of the parabola by solving $y = 0$.

Step-by-step Solution: 1. Set the vertex form equal to zero:

$$2(x-3)^2 - 8 = 0$$

2. Add 8 to both sides and divide by 2:

$$2(x-3)^2 = 8$$

$$(x-3)^2 = 4$$

3. Take the square root of both sides, remembering both positive and negative roots:

$$x - 3 = \pm\sqrt{4}$$

$$x - 3 = \pm 2$$

4. Solve for x: * **Case 1:** $x - 3 = 2 \implies x = 5$ * **Case 2:** $x - 3 = -2 \implies x = 1$ * The roots are $x = 1$ and $x = 5$.

- **Marking Rubric (2 Marks total):**
  - **1 Mark**: Setting up the equation and isolating $(x-3)^2 = 4$.
  - **1 Mark**: Finding both correct roots ($x = 1$ and $x = 5$).

Question 2

Solve the quadratic inequality $x^2 - 5x - 6 < 0$.

Step-by-step Solution: 1. **Factor the quadratic expression** to find the critical boundary points:

$$(x-6)(x+1) < 0$$

2. **Identify critical points** where the expression equals zero:

$$x = 6 \quad \text{and} \quad x = -1$$

3. **Analyze the intervals** created by these points ($x < -1$, $-1 < x < 6$, and $x > 6$): * Since the quadratic graph $y = x^2 - 5x - 6$ is a parabola opening upwards (coefficient of x^2 is positive), the graph lies below the x-axis ($y < 0$) *between* the roots. * Alternatively, we can use test values: * For $x = -2$ (Interval $x < -1$): $(-2-6)(-2+1) = (-8)(-1) = 8 > 0$ (does not satisfy inequality). * For $x = 0$ (Interval $-1 < x < 6$): $(0-6)(0+1) = -6 < 0$ (satisfies

inequality). * For $x = 7$ (Interval $x > 6$): $(7-6)(7+1) = 8 > 0$ (does not satisfy inequality). 4. **State the solution**: * The inequality is satisfied for values of x strictly between -1 and 6. * In interval notation, the solution is: $(-1, 6)$.

- **Marking Rubric (4 Marks total):**
  - **1 Mark**: Correct factorization $(x-6)(x+1)$.
  - **1 Mark**: Correctly identifying boundary values $x = -1$ and $x = 6$.
  - **1 Mark**: Logical testing of intervals (algebraic tests or graphical reasoning).
  - **1 Mark**: Correct solution written in interval notation $(-1, 6)$ or inequality form $-1 < x < 6$.

Topic 2: Log Equations

Question 3

Solve the logarithmic equation: $\log_2(x) + \log_2(x-2) = 3$.

Step-by-step Solution: 1. **Apply the product rule of logarithms** ($\log_b(A) + \log_b(B) = \log_b(A \cdot B)$):

$$\log_2[x(x-2)] = 3$$

2. **Convert the logarithmic equation to exponential form** ($y = \log_b(x) \iff b^y = x$):

$$x(x-2) = 2^3$$

$$x^2 - 2x = 8$$

3. **Rearrange into standard quadratic form** ($ax^2 + bx + c = 0$):

$$x^2 - 2x - 8 = 0$$

4. **Factor the quadratic equation**:

$$(x-4)(x+2) = 0$$

* Giving potential solutions: $x = 4$ and $x = -2$. 5. **Check for extraneous solutions** (the argument of a logarithm must be strictly positive): * For $x = 4$: * $\log_2(4)$ is valid (argument $4 > 0$). * $\log_2(4-2) = \log_2(2)$ is valid (argument $2 > 0$). * For $x = -2$: * $\log_2(-2)$ is invalid (argument $-2 \leq 0$). Thus, $x = -2$ is extraneous. * The only valid solution is $x = 4$.

- **Marking Rubric (4 Marks total):**
  - **1 Mark**: Correctly combining logs into $\log_2(x(x-2))$.
  - **1 Mark**: Converting to exponential form $x^2 - 2x = 8$.
  - **1 Mark**: Solving the quadratic equation to get $x = 4$ and $x = -2$.
  - **1 Mark**: Explicitly rejecting $x = -2$ due to domain constraints and stating final answer $x = 4$.

Question 4

Using the laws of logarithms, solve: $2\log_3(x) - \log_3(x+6) = 1$.

Step-by-step Solution: 1. **Apply the power rule** to the first term ($k\log_b(A) = \log_b(A^k)$):

$$\log_3(x^2) - \log_3(x+6) = 1$$

2. **Apply the quotient rule** ($\log_b(A) - \log_b(B) = \log_b(\frac{A}{B})$):

$$\log_3\left(\frac{x^2}{x+6}\right) = 1$$

3. **Convert to exponential form**:

$$\frac{x^2}{x+6} = 3^1$$

$$\frac{x^2}{x+6} = 3$$

4. **Solve the resulting equation**:

$$x^2 = 3(x+6)$$

$$x^2 = 3x + 18$$

$$x^2 - 3x - 18 = 0$$

5. **Factor the quadratic**:

$$(x-6)(x+3) = 0$$

* Giving potential solutions: $x = 6$ and $x = -3$. 6. **Check domain constraints**: * For $x = 6$: Both $\log_3(6)$ and $\log_3(12)$ are defined (valid). * For $x = -3$: $\log_3(-3)$ is undefined (invalid). * Therefore, the only valid solution is $x = 6$.

- **Marking Rubric (4 Marks total):**
  - **1 Mark**: Applying the power rule to write $\log_3(x^2)$.
  - **1 Mark**: Applying the quotient rule to obtain $\log_3(\frac{x^2}{x+6}) = 1$.
  - **1 Mark**: Converting to a quadratic equation $x^2 - 3x - 18 = 0$ and finding $x = 6$ and $x = -3$.
  - **1 Mark**: Rejecting the extraneous root $x = -3$ and stating final solution $x = 6$.

Topic 3: Trigonometry

Question 5

Triangle PAB with $P = 65^\circ$, $PA = 8\text{ m}$, and $PB = 12\text{ m}$.

Part a) Calculate the distance from A to B (to 3 s.f.).

Step-by-step Solution: 1. Since we know two sides and the included angle (SAS), we use the **Cosine Rule**:

$$c^2 = a^2 + b^2 - 2ab\cos(C)$$

Let $c = AB$, $a = 12\text{ m}$ (side opposite A), $b = 8\text{ m}$ (side opposite B), and $C = 65^\circ$. 2. Substitute the values:

$$AB^2 = 12^2 + 8^2 - 2(12)(8)\cos(65^\circ)$$

$$AB^2 = 144 + 64 - 192\cos(65^\circ)$$

$$AB^2 = 208 - 192(0.422618)$$

$$AB^2 \approx 208 - 81.1427$$

$$AB^2 \approx 126.8573$$

3. Take the square root:

$$AB = \sqrt{126.8573} \approx 11.263\text{ m}$$

4. Round to 3 significant figures: $AB \approx 11.3\text{ m}$.

- **Marking Rubric (3 Marks total):**
  - **1 Mark**: Selecting and stating the correct Cosine Rule formula.
  - **1 Mark**: Correct substitution of values into the formula.
  - **1 Mark**: Correct calculation and rounding to 11.3 m (with units).

Part b) Find the area of the triangular piece of land PAB (to 1 d.p.).

Step-by-step Solution: 1. Use the **sine area formula** ($\text{Area} = \frac{1}{2}ab\sin(C)$):

$$\text{Area} = \frac{1}{2} \cdot PB \cdot PA \cdot \sin(65^\circ)$$

2. Substitute the values:

$$\text{Area} = \frac{1}{2} \cdot 12 \cdot 8 \cdot \sin(65°)$$

$$\text{Area} = 48 \cdot \sin(65°)$$

$$\text{Area} \approx 48 \cdot 0.906308$$

$$\text{Area} \approx 43.5028 \text{ m}^2$$

3. Round to 1 decimal place: $\text{Area} \approx 43.5 \text{ m}^2$.

- **Marking Rubric (2 Marks total):**
  - **1 Mark**: Stating the correct area formula and substituting values.
  - **1 Mark**: Correct calculations and rounding to 43.5 m^2 (with units).

Question 6

Part a) Solve $2\sin(2x - \frac{\pi}{3}) = \sqrt{3}$ for $0 \leq x \leq \pi$.

Step-by-step Solution: 1. **Isolate the trigonometric term**:

$$\sin\left(2x - \frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$$

2. **Find the range for the compound angle $u = 2x - \frac{\pi}{3}$**: Since $0 \leq x \leq \pi \implies 0 \leq 2x \leq 2\pi$, subtracting $\frac{\pi}{3}$ gives:

$$-\frac{\pi}{3} \leq u \leq \frac{5\pi}{3}$$

3. **Find principal values** of u in this range where $\sin(u) = \frac{\sqrt{3}}{2}$: * In Quadrant I: $u = \frac{\pi}{3}$ * In Quadrant II: $u = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$ 4. **Solve for x in each case**: * **Case 1:** $2x - \frac{\pi}{3} = \frac{\pi}{3} \implies 2x = \frac{2\pi}{3} \implies x = \frac{\pi}{3}$ * **Case 2:** $2x - \frac{\pi}{3} = \frac{2\pi}{3} \implies 2x = \pi \implies x = \frac{\pi}{2}$ * Both solutions lie in the interval $[0, \pi]$. * Solutions: $x = \frac{\pi}{3}, \frac{\pi}{2}$.

- **Marking Rubric (3 Marks total):**
  - **1 Mark**: Isolating the sine term to obtain $\sin(2x - \frac{\pi}{3}) = \frac{\sqrt{3}}{2}$.
  - **1 Mark**: Determining the two angles for the compound angle ($\frac{\pi}{3}$ and $\frac{2\pi}{3}$).
  - **1 Mark**: Finding both correct values for x within the interval.

Part b) Solve $2\cos^2(x) - \sin(x) - 1 = 0$ for $0 \leq x \leq 2\pi$.

Step-by-step Solution: 1. **Use the Pythagorean Identity** $\cos^2(x) = 1 - \sin^2(x)$ to convert the equation to a quadratic in terms of $\sin(x)$:

$$2(1 - \sin^2(x)) - \sin(x) - 1 = 0$$

$$2 - 2\sin^2(x) - \sin(x) - 1 = 0$$

$$-2\sin^2(x) - \sin(x) + 1 = 0$$

2. **Multiply by** -1 to get standard form:

$$2\sin^2(x) + \sin(x) - 1 = 0$$

3. **Factor the quadratic expression** (let $s = \sin(x)$ so $2s^2 + s - 1 = 0$):

$$(2\sin(x) - 1)(\sin(x) + 1) = 0$$

4. **Solve the separate equations**: * **Equation 1:** $2\sin(x) - 1 = 0 \implies \sin(x) = \frac{1}{2}$ * In the interval $0 \leq x \leq 2\pi$: $x = \frac{\pi}{6}$ and $x = \frac{5\pi}{6}$. * **Equation 2:** $\sin(x) + 1 = 0 \implies \sin(x) = -1$ * In the interval $0 \leq x \leq 2\pi$: $x = \frac{3\pi}{2}$. 5. Solutions: $x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$.

- **Marking Rubric (4 Marks total):**
  - **1 Mark**: Using the identity to convert the equation to terms of $\sin(x)$.
  - **1 Mark**: Correctly factoring the quadratic equation into $(2\sin(x) - 1)(\sin(x) + 1) = 0$.
  - **1 Mark**: Solving $\sin(x) = \frac{1}{2}$ to get $x = \frac{\pi}{6}, \frac{5\pi}{6}$.
  - **1 Mark**: Solving $\sin(x) = -1$ to get $x = \frac{3\pi}{2}$.

Topic 4: Probability

Question 7

Total students $N = 50$, Math (M) = 30, Physics (P) = 25, Both ($M \cap P$) = 10.

Part a) Construct a Venn diagram.

Step-by-step Solution: 1. Start with the intersection: $M \cap P = 10$. 2. Calculate the number of students who study *only* Mathematics:

$$\text{Only Math} = N(M) - N(M \cap P) = 30 - 10 = 20$$

3. Calculate the number of students who study *only* Physics:

$$\text{Only Physics} = N(P) - N(M \cap P) = 25 - 10 = 15$$

4. Calculate the number of students who study *neither*:

$$\text{Neither} = 50 - (\text{Only Math} + \text{Intersection} + \text{Only Physics}) = 50 - (20 + 10 + 15) = 5$$

5. Draw two overlapping circles labeled M and P inside a rectangular box U: * M circle (exclusive area): 20 * Overlap area: 10 * P circle (exclusive area): 15 * Box area (outside circles): 5

- **Marking Rubric (3 Marks total):**
  - **1 Mark**: Putting 10 in the intersection.
  - **1 Mark**: Calculating and placing 20 and 15 in the respective outer circles.
  - **1 Mark**: Placing 5 in the bounding box outside the circles.

Part b) Find the probability they study Mathematics, given they study Physics.

Step-by-step Solution: 1. Use the **conditional probability formula** ($P(A|B) = \frac{P(A \cap B)}{P(B)}$):

$$P(M|P) = \frac{N(M \cap P)}{N(P)}$$

2. Substitute the values:

$$P(M|P) = \frac{10}{25} = \frac{2}{5} = 0.4$$

- **Marking Rubric (2 Marks total):**
  - **1 Mark**: Showing correct probability definition/fraction $\frac{10}{25}$.
  - **1 Mark**: Stating the correct final answer (0.4 or $\frac{2}{5}$).

Part c) Find the probability they do not study Mathematics, given they study Physics.

Step-by-step Solution: 1. The probability of the complement event under the same condition is:

$$P(M'|P) = 1 - P(M|P) = 1 - 0.4 = 0.6$$

2. Alternatively, read directly from the Venn diagram:

$$P(M'|P) = \frac{\text{Only Physics}}{P} = \frac{15}{25} = \frac{3}{5} = 0.6$$

- **Marking Rubric (2 Marks total):**
  - **1 Mark**: Stating relationship $1 - P(M|P)$ or fraction $\frac{15}{25}$.
  - **1 Mark**: Finding the final correct value (0.6 or $\frac{3}{5}$).

Question 8

Box with 5 red, 3 green balls. Two draws without replacement.

Part a) Draw a tree diagram.

Step-by-step Solution: 1. **First Draw** (8 balls in total): * Branch 1: Red (R_1) with probability $\frac{5}{8}$ * Branch 2: Green (G_1) with probability $\frac{3}{8}$ 2. **Second Draw** (7 balls remaining): * If first was Red (R_1, leaves 4 Red, 3 Green): * Branch 1a: Red (R_2) with probability $\frac{4}{7}$ * Branch 1b: Green (G_2) with probability $\frac{3}{7}$ * If first was Green (G_1, leaves 5 Red, 2 Green): * Branch 2a: Red (R_2) with probability $\frac{5}{7}$ * Branch 2b: Green (G_2) with probability $\frac{2}{7}$

- **Marking Rubric (3 Marks total):**
  - **1 Mark**: Correct structure (two branches splitting into two).
  - **1 Mark**: Correct probabilities for the first draw branches.
  - **1 Mark**: Correct conditional probabilities for the second draw branches.

Part b) Calculate the probability that the two balls drawn are of different colors.

Step-by-step Solution: 1. The outcome "different colors" corresponds to two mutually exclusive paths: * **Path 1:** Red then Green ($R_1 \cap G_2$)

$$P(R_1 \cap G_2) = \frac{5}{8} \times \frac{3}{7} = \frac{15}{56}$$

* **Path 2:** Green then Red ($G_1 \cap R_2$)

$$P(G_1 \cap R_2) = \frac{3}{8} \times \frac{5}{7} = \frac{15}{56}$$

2. Add the probabilities of the two paths:

$$P(\text{Different Colors}) = P(R_1 \cap G_2) + P(G_1 \cap R_2)$$

$$P(\text{Different Colors}) = \frac{15}{56} + \frac{15}{56} = \frac{30}{56} = \frac{15}{28} \approx 0.536$$

- **Marking Rubric (3 Marks total):**
    - **1 Mark**: Correct calculations for $P(RG) = \frac{15}{56}$.
    - **1 Mark**: Correct calculations for $P(GR) = \frac{15}{56}$.
    - **1 Mark**: Summing the paths to get the correct final answer $\frac{15}{28}$ (or ≈ 0.536).

Topic 5: Vectors in 2D

Question 9

$\vec{u} = \begin{pmatrix} 4 & -3 \end{pmatrix}, \vec{v} = \begin{pmatrix} -2 & 5 \end{pmatrix}$.

Part a) Calculate the resultant vector $\vec{w} = 3\vec{u} + 2\vec{v}$.

Step-by-step Solution: 1. Perform scalar multiplication on each vector:

$$3\vec{u} = 3\begin{pmatrix} 4 & -3 \end{pmatrix} = \begin{pmatrix} 12 & -9 \end{pmatrix}$$
$$2\vec{v} = 2\begin{pmatrix} -2 & 5 \end{pmatrix} = \begin{pmatrix} -4 & 10 \end{pmatrix}$$

2. Add the component values:

$$\vec{w} = \begin{pmatrix} 12 & -9 \end{pmatrix} + \begin{pmatrix} -4 & 10 \end{pmatrix} = \begin{pmatrix} 12 + (-4) & -9 + 10 \end{pmatrix} = \begin{pmatrix} 8 & 1 \end{pmatrix}$$

- **Marking Rubric (2 Marks total):**
  - **1 Mark**: Showing correct scalar multiplication.
  - **1 Mark**: Correct addition to find $\vec{w} = \begin{pmatrix} 8 & 1 \end{pmatrix}$.

Part b) Calculate the magnitude of vector $\vec{u}$ and vector $\vec{w}$.

Step-by-step Solution: 1. The magnitude of a vector $\vec{a} = \begin{pmatrix} x & y \end{pmatrix}$ is given by $|\vec{a}| = \sqrt{x^2 + y^2}$.
2. Calculate $|\vec{u}|$:

$$|\vec{u}| = \sqrt{4^2 + (-3)^2} = \sqrt{16 + 9} = \sqrt{25} = 5$$

3. Calculate $|\vec{w}|$:

$$|\vec{w}| = \sqrt{8^2 + 1^2} = \sqrt{64 + 1} = \sqrt{65} \approx 8.06$$

- **Marking Rubric (3 Marks total):**
  - **1 Mark**: Stating the correct formula for magnitude (Pythagoras).
  - **1 Mark**: Finding $|\vec{u}| = 5$.
  - **1 Mark**: Finding $|\vec{w}| = \sqrt{65} \approx 8.06$.

Question 10

Vectors $\vec{a} = 3\vec{i} + 4\vec{j}$ and $\vec{b} = 12\vec{i} - 5\vec{j}$.

Part a) Calculate the scalar product (dot product) $\vec{a} \cdot \vec{b}$.

Step-by-step Solution: 1. Use components multiplication formula $\vec{a} \cdot \vec{b} = a_x b_x + a_y b_y$:

$$\vec{a} \cdot \vec{b} = (3)(12) + (4)(-5)$$

$$\vec{a} \cdot \vec{b} = 36 - 20 = 16$$

- **Marking Rubric (2 Marks total):**
  - **1 Mark**: Showing step $(3)(12) + (4)(-5)$.
  - **1 Mark**: Correct final scalar product value (16).

Part b) Find the angle θ between vectors $\vec{a}$ and $\vec{b}$ (to nearest degree).

Step-by-step Solution: 1. Use the geometric definition of the scalar product:

$$\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos(\theta) \implies \cos(\theta) = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|}$$

2. Calculate the magnitudes of vectors $\vec{a}$ and $\vec{b}$:

$$|\vec{a}| = \sqrt{3^2 + 4^2} = \sqrt{25} = 5$$

$$|\vec{b}| = \sqrt{12^2 + (-5)^2} = \sqrt{169} = 13$$

3. Substitute into the cosine formula:

$$\cos(\theta) = \frac{16}{5 \times 13} = \frac{16}{65} \approx 0.24615$$

4. Take the inverse cosine:

$$\theta = \arccos(0.24615) \approx 75.75^\circ$$

5. Round to the nearest degree: $\theta \approx 76^\circ$.

- **Marking Rubric (3 Marks total):**
  - **1 Mark**: Finding magnitudes $|\vec{a}| = 5$ and $|\vec{b}| = 13$.
  - **1 Mark**: Substituting values into equation to find $\cos(\theta) = \frac{16}{65}$.
  - **1 Mark**: Correct inverse calculation and rounding to 76°.

Topic 6: 3D Shapes

Question 11

Cuboid with $AB = 3$ cm, $BC = 4$ cm, and height $CG = 12$ cm.

Part a) Calculate the length of the base diagonal, AC.

Step-by-step Solution: 1. The base of the cuboid is a rectangle $ABCD$ with sides $AB = 3$ cm and $BC = 4$ cm. The diagonal AC forms a right-angled triangle ABC. 2. Apply Pythagoras theorem in 2D:

$$AC = \sqrt{AB^2 + BC^2}$$

$$AC = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5 \text{ cm}$$

- **Marking Rubric (2 Marks total):**
  - **1 Mark**: Setting up the Pythagorean relationship $AC^2 = 3^2 + 4^2$.
  - **1 Mark**: Correct calculation to get 5 cm (with units).

Part b) Calculate the length of the space diagonal of the cuboid, AG.

Step-by-step Solution: 1. The space diagonal AG forms a right-angled triangle ACG where $AC = 5$ cm (base diagonal) and $CG = 12$ cm (vertical height). 2. Apply Pythagoras theorem:

$$AG = \sqrt{AC^2 + CG^2}$$

$$AG = \sqrt{5^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169} = 13 \text{ cm}$$

(Note: Alternatively, we can use the direct 3D Pythagoras formula $AG = \sqrt{x^2 + y^2 + z^2} = \sqrt{3^2 + 4^2 + 12^2} = 13 \text{ cm}$).

- **Marking Rubric (2 Marks total):**
  - **1 Mark**: Setting up the Pythagorean equation with either 2D intermediate or direct 3D variables.
  - **1 Mark**: Correct calculation to get 13 cm (with units).

Question 12

Part a) Cylinder total surface area ($r = 3$ cm, $h = 8$ cm) in terms of π.

Step-by-step Solution: 1. State the formula for the **total surface area of a closed cylinder**:

$$\text{Area} = 2\pi r^2 + 2\pi r h$$

* $2\pi r^2$ represents the area of the top and bottom circular bases. * $2\pi rh$ represents the curved surface area (circumference multiplied by height). 2. Substitute the given values $r = 3 \text{ cm}$ and $h = 8 \text{ cm}$:

$$\text{Area} = 2\pi(3)^2 + 2\pi(3)(8)$$

$$\text{Area} = 2\pi(9) + 2\pi(24)$$

$$\text{Area} = 18\pi + 48\pi$$

3. Combine terms (leaving in terms of π as requested):

$$\text{Area} = 66\pi \text{ cm}^2$$

- **Marking Rubric (3 Marks total):**
  - **1 Mark**: Stating the correct formula for total surface area.
  - **1 Mark**: Correct substitution of values.
  - **1 Mark**: Correct simplification to $66\pi \text{ cm}^2$ (with units).

Part b) Volume of a right pyramid (square base side 6 cm, vertical height 10 cm).

Step-by-step Solution: 1. State the formula for the **volume of a pyramid**:

$$\text{Volume} = \frac{1}{3} \times \text{Base Area} \times \text{Height}$$

2. Calculate the base area of the square base ($s = 6 \text{ cm}$):

$$\text{Base Area} = s^2 = 6^2 = 36 \text{ cm}^2$$

3. Substitute the base area and vertical height ($h = 10 \text{ cm}$) into the volume formula:

$$\text{Volume} = \frac{1}{3} \times 36 \times 10$$

$$\text{Volume} = 12 \times 10 = 120 \text{ cm}^3$$

- **Marking Rubric (2 Marks total):**
  - **1 Mark**: Calculating correct base area (36 cm^2) and stating the volume formula.
  - **1 Mark**: Correct final volume calculation of 120 cm^3 (with units).